# REPORT 002: EPOCH 2: PORT-INDEPENDENT USB/ALSA HARDWARE FINGERPRINTING & SIMULTANEOUS MULTI-DEVICE MIDI MANAGER

2026-11-04 00:08:30 | Epoch 2 of 20: src/hardware.rs (sysfs, /dev/snd, /proc/asound fingerprinting) & src/midi_io.rs (MultiDeviceManager & hot-plug persistence)

Operator: Gemini 4 Argon (Medium) · Follows: report-001

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## Prompt

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Epoch 2 of 20: Implement src/hardware.rs and src/midi_io.rs for simultaneous
↪ multi-device MIDI controller management and USB port-independent hardware
↪ identity persistence. Inspect Linux /sys/class/sound/card*/ (id,
↪ device/../idVendor, device/../idProduct, device/../serial,
↪ device/../manufacturer, device/../product), /dev/snd/by-id/, /dev/snd/by-path/,
↪ and /proc/asound/seq/clients (plus aconnect -l / amidi -l fallback) to compute a
↪ stable HardwareFingerprint (vendor_id:product_id:serial:card_name, such as
↪ 0944:0117 for the currently connected Korg nanoKONTROL2 on card2 / ALSA client
↪ 24:0). Build the MultiDeviceManager that connects multiple controllers
↪ simultaneously, remembers each device's hardware identity and assigned
↪ preset/role in persistent config, and automatically detects hot-plug removal and
↪ reconnection-even when plugged into a different physical USB port where the ALSA
↪ client number (e.g. 24:0 -> 28:0) or USB bus path changes. Add unit and
↪ integration tests including read-only verification against the connected Korg
↪ nanoKONTROL2 (if present) and simulated multi-port USB re-plug scenarios (with
↪ CARGO_BUILD_JOBS=2), git commit all changes, and compile reports/report-002.html
↪ from reports/report-002.md via
↪ /home/renick/claude/html-report-generator/build-report.sh --pdf and commit.
```

## TL;DR

Implemented `src/hardware.rs` and `src/midi_io.rs` for simultaneous multi-device MIDI controller management and USB port-independent hardware identity persistence. `src/hardware.rs` inspects Linux `/sys/class/sound/card*/` (`id`, `device/../idVendor`, `device/../idProduct`, `device/../serial`, `device/../manufacturer`, `device/../product`), `/dev/snd/by-id/`, `/dev/snd/by-path/`, and `/proc/asound/seq/clients` (with `aconnect -l` and `amidi -l` enrichment) to compute a stable `HardwareFingerprint` (`vendor_id:product_id:serial:card_name`, verified live against the connected Korg nanoKONTROL2 `0944:0117::nanoKONTROL2` on `card2` / ALSA `24:0` and Behringer UMC1820 `1397:0503:B8AAF8D3:UMC1820` on `card0` / ALSA `16:0`). `src/midi_io.rs` implements `MultiDeviceManager` with persistent profile storage, hot-plug detection across changing USB ports (`24:0` -> `32:0`), and automatic 28-LED resynchronization, verified by 22 passing automated tests (`17` Rust unit/integration tests + `5` Node.js SDK tests).

## 1. Plain-Language Overview & Epoch 2 Goals

When a musician plugs a USB MIDI controller into a Linux computer, the operating system's audio and MIDI subsystem (**ALSA**, the Advanced Linux Sound Architecture) assigns it a temporary sound card number (like `card2`) and a temporary MIDI client number (like `24:0`). If the musician unplugs that controller during a rehearsal and plugs it into a different USB socket on the laptop or USB hub, Linux often gives it a brand-new card number (`card3` or `card4`), a different ALSA client number (`28:0` or `32:0`), and a different physical bus path (`/dev/snd/by-path/pci-...-usb-0:2.4:1.0`).

In naive MIDI programs, changing the USB socket breaks the connection: the software forgets which controller is which, loses the loaded mapping preset, and leaves all the physical button lights (LEDs) dark.

In **Epoch 2 of 20**, we solved this problem permanently by building two core Rust modules:

1. **Port-Independent Hardware Fingerprinting (`src/hardware.rs`)**: Reads the hardware's permanent USB identity directly from the Linux kernel (`/sys/class/sound/card*/`, `/dev/snd/by-id/`, `/dev/snd/by-path/`, and `/proc/asound/seq/clients`, plus `aconnect -l` and `amidi -l`) to create a `HardwareFingerprint` (`vendor_id:product_id:serial:card_name`). Even if the USB socket, card number, and ALSA port number all change, the hardware fingerprint (`0944:0117::nanoKONTROL2` for the Korg nanoKONTROL2) stays 100% identical.
2. **Simultaneous Multi-Device Manager (`src/midi_io.rs`)**: Manages multiple MIDI controllers at the same time (`MultiDeviceManager`), saves each controller's assigned preset, role (`DeviceRole`), and LED states in `PersistentHardwareConfig`, and automatically detects when a controller is unplugged and plugged back into any USB port, restoring its preset and queuing the exact MIDI messages to relight all 28 hardware LEDs.

## Plain-English Concepts Mapped to Code Components

| Plain-English Concept | What It Does for the Musician | Exact Code Component |
|---|---|---|
| Permanent Hardware Fingerprint | Identifies a controller by its USB Vendor/Product ID, Serial, and Card ID (`0944:0117::nanoKONTROL2`) instead of a temporary port number | `src/hardware.rs` (`HardwareFingerprint`, `HardwareFingerprint::matches_replugged`) |
| Linux Sysfs Sound Card Scanner | Reads `/sys/class/sound/card*/` and `/dev/snd/by-{id,path}` symlinks | `src/hardware.rs` (`SysfsSoundCard`, `scan_sysfs_sound_cards`, `scan_dev_snd_symlinks`) |
| ALSA Sequencer & RawMIDI Discovery | Parses `/proc/asound/seq/clients`, `aconnect -l`, and `amidi -l` | `src/hardware.rs` (`AlsaSeqClient`, `AlsaRawMidiEntry`, `parse_proc_asound_seq_clients`) |
| Unified Device Correlator | Merges sysfs USB identity with live ALSA `24:0` and `hw:2,0,0` addresses | `src/hardware.rs` (`DiscoveredMidiDevice`, `correlate_hardware_devices`, `discover_connected_midi_hardware`) |
| Simultaneous Multi-Device Manager | Connects multiple controllers at once, tracks hot-plug events, and persists settings | `src/midi_io.rs` (`MultiDeviceManager`, `ManagedDeviceSession`, `HotplugEvent`) |
| Automatic LED Resynchronization | Generates the 28 CC or 24 Note MIDI packets to restore button LEDs on reconnect | `src/midi_io.rs` (`ManagedDeviceSession::build_led_sync_packets`, `RawMidiPacket`) |

## 2. Phase 1: Linux Sysfs, '/dev/snd', and '/proc/asound' Hardware Discovery ('src/hardware.rs')

#### Correlating USB Descriptors with Dynamic ALSA Ports

`src/hardware.rs` inspects four layers of Linux kernel and ALSA state in read-only mode:

- **`/sys/class/sound/card*/`**: Reads `id` (e.g. `nanoKONTROL2`), checks for `midiC*D*` / `dmmidi*` child nodes, and reads USB device attributes (`idVendor`, `idProduct`, `serial`, `manufacturer`, `product`) from `device/...`
- **`/dev/snd/by-id/` and `/dev/snd/by-path/`**: Maps each `../controlCN` symlink target back to `card_index` so every `SysfsSoundCard` knows both its stable USB symlink (`usb-KORG_INC._nanoKONTROL2-00`) and its current physical USB socket (`pci-0000:00:14.0-usb-0:5.3:1.0`).
- **`/proc/asound/seq/clients` and `aconnect -l` / `amidi -l`**: Extracts the active ALSA Sequencer client and port (`24:0`, `"nanoKONTROL2 _ CTRL"`) and RawMIDI address (`hw:2,0,0`, `"nanoKONTROL2 _ SLIDER/KNOB"`).

> #### Preventing USB Host Controller Serial Bleed When Device Lacks a USB Serial Descriptor
>
> **Symptom** During our first read-only test against the live Korg nanoKONTROL2 on `card2`, `nk2.fingerprint.canonical_key()` returned `"0944:0117:0000:00:14.0:nanoKONTROL2"` instead of `"0944:0117::nanoKONTROL2"`.
>
> **Root cause** Unlike the Behringer UMC1820 (which exposes `/sys/class/sound/card0/device/../serial = B8AAF8D3`), the Korg nanoKONTROL2 does not have a `serial` file in its USB device directory (`1-5.3`). When `read_usb_attr` walked up 4 parent directories looking for `serial`, it climbed out of the USB device into the PCI USB host controller (`usb1`), whose `serial` file contains the PCI bus address `0000:00:14.0`.
>
> **Fix** Updated `read_usb_attr` in `src/hardware.rs` to anchor strictly on the directory containing `idVendor` and `idProduct` (`card_dir/device/..` or `card_dir/device`) and never walk higher up the sysfs tree.
>
> **Rationale** Anchoring on the USB device directory that owns `idVendor` and `idProduct` guarantees that devices with a true hardware serial number (like `UMC1820` -> `B8AAF8D3`) report their serial, while devices without a USB serial descriptor (like `nanoKONTROL2`) cleanly report `""` instead of inheriting the PCI host controller's address.

## Live Host Hardware Discovery Results ('discover_connected_midi_hardware()')

| Discovered Device | 'HardwareFingerprint' | Matched Definition | Sysfs Card & '/dev/snd/by-id' | ALSA Seq & RawMIDI | Physical USB Path ('/dev/snd/by-path') |
|---|---|---|---|---|---|
| **Korg nanoKONTROL2** | `0944:0117::nanoKONTROL2` | `KORG nanoKONTROL2` | `card2` (`usb-KORG_INC._nanoKONTROL2-00`) | `24:0` (`nanoKONTROL2 _ CTRL`), `hw:2,0,0` | `pci-0000:00:14.0-usb-0:5.3:1.0` |
| **Behringer UMC1820** | `1397:0503:B8AAF8D3` | `BEHRINGER UMC1820 (Auxiliary)` | `card0` (`usb-BEHRINGER_UMC1820_B8AAF8D3-00`) | `16:0` (`UMC1820 MIDI 1`), `hw:0,0,0` | `pci-0000:00:14.0-usb-0:6:1.0` |

## 3. Phase 2: Simultaneous Multi-Device Manager & Hot-Plug Persistence (‘src/midi_io.rs‘)

**Multi-Controller Sessions, USB Socket Migration, and LED Resynchronization**

`src/midi_io.rs` provides `MultiDeviceManager`, which maintains active and remembered `ManagedDeviceSession` entries alongside a JSON-serializable `PersistentHardwareConfig`:

- **Simultaneous Multi-Device Management**: Connects and tracks multiple controllers at the same time (for example, a Korg `nanoKONTROL2` on `24:0`, an Akai `MIDI Mix` on `28:0`, and a `UMC1820` on `16:0`), each with its own independent `DeviceRole`, `active_preset`, `control_values`, and `led_states`.
- **Hot-Plug Removal & USB Port Migration (`reconcile_discovered_devices`)**: Emits `HotplugEvent::ConnectedNew` when a device is seen for the first time, `HotplugEvent::Disconnected` when a connected controller disappears, and `HotplugEvent::ReconnectedRemembered` when a remembered device reappears—even if plugged into a different physical USB port where its ALSA address changes (e.g. `24:0` -> `32:0`), its card index changes (`card2` -> `card4`), and its `/dev/snd/by-path` changes (`usb-0:5.3:1.0` -> `usb-0:2.4:1.0`).
- **Automatic Hardware LED Resynchronization (`build_led_sync_packets`)**: Immediately upon reconnection, `MultiDeviceManager` restores the device’s `active_preset` and queues all 28 CC packets (for `nanokontrol2`) or 24 Note packets (for `akai-midimix`) to bring the physical controller’s LEDs back into sync with the software state.

## Tests

- `CARGO_BUILD_JOBS=2 cargo test` — 17 Rust unit and integration tests passing (`7` new tests in `hardware::tests` and `midi_io::tests` including live read-only Korg nanoKONTROL2 verification and simulated multi-port USB re-plug migration), 0 failures
- `CARGO_BUILD_JOBS=2 cargo run -- --check-init` — Verified live discovery of 2 connected hardware MIDI devices (`nanoKONTROL2` and `UMC1820`)
- `npm test` (`node --test js/*.test.js`) — 5 Node.js SDK unit tests passing, 0 failures

## QA

```
`22 tests passing (17 Rust + 5 Node.js)`

`Live hardware fingerprint verified: 0944:0117::nanoKONTROL2 (card2, ALSA 24:0, hw:2,0,0)`

`Simulated USB port migration (ALSA 24:0 -> 32:0) & 28-LED resync verified`

`0 warnings, 0 failures`
```

All Epoch 2 deliverables compiled cleanly with `CARGO_BUILD_JOBS=2` and passed all Rust and Node.js test suites in read-only hardware mode.

## Files touched

- ‘src/hardware.rs‘ — Implemented Linux sysfs (‘/sys/class/sound/card*/‘), ‘/dev/snd/by-id/‘, ‘/dev/snd/by-path/‘, ‘/proc/asound/seq/clients‘, ‘aconnect -l‘, and ‘amidi -l‘ discovery and ‘HardwareFingerprint‘ correlation
- ‘src/midi_io.rs‘ — Implemented ‘MultiDeviceManager‘, ‘ManagedDeviceSession‘, ‘PersistentHardwareConfig‘, ‘HotplugEvent‘, and ‘RawMidiPacket‘ LED synchronization

- ‘src/lib.rs‘ — Exported ‘hardware‘ and ‘midi_io‘ modules
- ‘src/main.rs‘ — Integrated ‘MultiDeviceManager::poll_host_hardware()‘ into ‘NkEditorApp‘ and ‘--check-init‘ CLI diagnostics
- ‘js/nkeditor-sdk.js‘ — Added ‘formatHardwareFingerprint‘ and ‘parseHardwareFingerprint‘ helpers to the companion Node.js SDK
- ‘js/nkeditor-sdk.test.js‘ — Added unit tests for ‘HardwareFingerprint‘ formatting and parsing

## Test coverage (22 total)

## QA verdict

**PASS**

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Report 002 | Generated 2026-11-04 00:08:30 | reports/report-002.html